

National Institute of Textile Engineering and Research (NITER)

Nayarhat, Savar, Dhaka

BSc. In CSE Level-1 Term-2 Class Test - 01

Course Code: CSE=1201

Course Name: Fundamentals of Programming

Time: 1 hour

Marks: 25

Answer any of the 5 questions following

1.	Describe briefly about break and continue statements with proper examples.	5
2.	<pre>int main() { int n = 20; if (n > 10) printf("A\n"); if (n > 15) printf("B\n"); else printf("C\n"); return 0; }</pre> i. What is output? ii. What change(s) would make the logical flow more consistent?	5
3.	i. <pre>int i = 0; while (i < 3) { for (int j = 0; j < 2; j++) { printf("%d%d ", i, j); } i++; }</pre> How many times will "11" be printed and why? ii. <pre>for (int i = 0; i < 3; i++) { for (int j = 0; j < 2; j++) { printf("%d%d ", i, j); } }</pre> How many times will "00" be printed and why?	5
4.	How the following program will be executed?	
	<pre>int num; int sum = 0;</pre> else if (num < 0) <pre>{</pre>	

	<pre> printf("Enter integers (0 to stop):\n"); while (1) { printf("Enter an integer: "); if (scanf("%d", &num) != 1) { printf("Invalid input. Exiting.\n"); return 1; } if (num == 0) { break; } } </pre>	<pre> continue; } else if (num >= 100) { break; } else { sum += num; } } printf("Sum of accepted numbers = %d\n", sum); return 0; } </pre>	
5.	How nested if works?		5
6.	Write a program that reads integers from the user until a negative number is entered. For each integer, If it's zero, break, if it's positive > 0, increment a counter. At the end, print the total count of positive numbers entered.		5
7.	Why do-while loop is different from for and while loop in syntax structure?		5

National Institute of Textile Engineering and Research (NITER)

1st Year 2nd Semester B.Sc. (1st Class Test) Examination-2024

Department of Computer Science and Engineering (CSE)

MATH 1204: Methods of Integration, Differential Equations and Series

Duration: 1 Hour

Full Marks: 25

Answer any Five (05) of the following questions.

1. Evaluate (i) $\int \sin^3 x \cos^2 x dx$ and (ii) $\int_0^1 \tan^{-1} x dx$. 5
2. Evaluate $\int_0^\pi \sin x dx$ by dividing the interval into eight strips using Simpson's rule. 5
3. Discuss improper integral of second kind with example. Examine the convergency and evaluate $\int_{-\infty}^{\infty} \frac{x}{x^4 + 1} dx$. 5
4. Find the length of one arch of the cycloid $x = r(\theta - \sin \theta)$, $y = r(1 - \cos \theta)$. 5
5. Find the surface area of the solid generated by revolving the parabola $y^2 = 4ax$ between $x = 0$ and $x = a$ about x-axis. 5
6. Discuss order of a Differential Equation and show that the differential equation of the circle tangent to the x-axis at the origin is $(x^2 - y^2)dy - 2xydx = 0$. 5

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National Institute of Textile Engineering and Research (NITER)
Department: CSE, Course Code: PHY-1203, Total Marks: 25, Time: 1 Hour

Section A: Answer Any One

(10)

3. For adiabatic process prove that $PV^\gamma = \text{constant}$, where symbols have their usual meaning.
4. Describe in detail each stage of the **Carnot cycle**, and derive the expression for the **net work done** per cycle.

Section B: Answer Any One

(06)

3. Prove that $C_p - C_v = R$, where symbols have their usual meaning.
4. Derive the equation for the **work done** during the expansion of a gas in an **adiabatic process**.

Section C: Answer Any One

(05)

3. A gas is initially at 47°C . Calculate the rise in its temperature if the pressure is suddenly doubled. Take $\gamma = 1.4$.
4. A Carnot engine absorbs 10^5 J of heat from a source at 627°C and rejects heat to a sink at 25°C . Calculate the amount of work done performed by the engine.

Section D: Answer Any Two

(2 × 2 = 4)

4. Write down the **First Law of Thermodynamics** for different thermodynamic processes.
5. Draw and explain the **P vs V diagrams** corresponding to different thermodynamic processes.
6. Define: Internal energy, Molar specific heat, Second law of thermodynamics.

1st Year 2nd Semester B.Sc. (1st Class Test) Examination-2024
Dept. of Computer Science and Engineering
National Institute of Textile Engineering and Research (NITER)

Course Code: CSE-1202

Course Name: Digital Logic Design (3 Credit)

Full Marks: 25

Time: 1 Hour

1. Simplify the following Boolean expressions to a minimum number of literals: 15
 - I. $AB'C' + A'BC + ABC' + AB'C$
 - II. $A'B'C' + A'BC + AB'C + ABC'$
 - III. $(A' + C')(A + B' + C')$
 - IV. $(X'Y' + Z)' + Z + XY + WZ$
 - V. $(A' + C)(A' + C')(A + B + C'D)$
2. Implement the Boolean function $F = xy + x'y' + y'z$ with OR and inverter gates. 5
3. Draw the logic diagram of the function $F = (AB + A'B')(CD' + C'D)$ without simplifying it. 5

Year-1 Semester-2 B.Sc. (1st In-course) Examination-2024
Dept. of Computer Science and Engineering (CSE)
National Institute of Textile Engineering and Research (NITER)
Course Code: ENG- 1205

Course Name: Developing English Language Skills (Credit: 2.0)

Full Marks: 25

Time: 1 Hour

Answer the following questions:

1. Write a sample CV with a cover letter. 15
3. Fill in the following text with appropriate prepositions: 10

(a)_____ a beautiful spring day, a curious squirrel was hopping (b) _____ search of food. Due (c) _____ the blossoming flowers, the garden was full (d) _____ life. Suddenly, it spotted a nut (e) _____ the grass. When it dashed (f) _____ it, it realized the nut was (g) _____ the base (h) _____ a tall tree. As it wondered how to reach the nut, it saw a rope hanging (i) _____. It started climbing (j) _____ the rope and in no time, it reached the nut. The squirrel happily munched on the nut and hopped away.